

Vinayak Chaturvedi

✉ vinayakchaturvedi044@gmail.com

☎ +91-9601669717

Projects

Data Job Market Analysis — Python, PostgreSQL, Power BI, Pandas, Matplotlib

- Processed and analyzed **750,000+** job market records using **Python (Pandas)** and **PostgreSQL**, uncovering hiring trends, salary benchmarks, and in-demand skills across major data-focused roles.
- Performed data cleaning, transformation, and exploratory analysis to identify high-demand and high-paying skills, enabling data-driven insights into career opportunities within the data analytics industry.
- Designed an interactive **Power BI dashboard** featuring KPIs, salary benchmarks, skill popularity metrics, and job market trends, delivering actionable insights through a centralized analytics platform.

Consumer Market Trends Analysis — Python, PostgreSQL, Power BI, Pandas

- Evaluated customer purchasing trends from **3,900+** retail transactions, uncovering revenue drivers, demographic spending patterns, and factors influencing customer retention.
- Conducted end-to-end analysis using **Python** and **PostgreSQL**, generating insights on subscription behavior, discount performance, customer loyalty, and product demand to support strategic decision-making.
- Built a dynamic **Power BI dashboard** presenting KPIs, customer segments, and sales trends, enabling data-driven recommendations for marketing, inventory planning, and customer engagement.

Library Operations Analysis — PostgreSQL, Advanced SQL

- Engineered a relational database system for managing library operations, including data modeling, table creation, record population, data maintenance, and end-to-end CRUD functionality across multiple entities.
- Automated circulation workflows through **PL/pgSQL stored procedures**, reducing manual intervention in book issuance and return tracking while ensuring consistency of inventory records.
- Leveraged advanced SQL techniques including **CTEs, Joins, Subqueries, CTAS, Aggregations, and Stored Procedures** to address real-world business scenarios such as overdue book monitoring, branch revenue analysis, member engagement tracking, and operational performance reporting.

Experience & Achievements

- Earned the **Data Analysis with Python** certification from freeCodeCamp, gaining proficiency in Python, NumPy, Pandas, Matplotlib, and Seaborn through projects including data cleaning, demographic analysis, medical data visualization, time-series analysis, and sea-level prediction.
- Earned the **Marketing Analysis** certification from NPTEL (IIT Kharagpur), developing skills in market research, consumer insights, and data-driven business decision making.
- Led a team to develop a **Capstone Group Management System** that streamlined group creation, auto-matched supervisor allocation, and implemented unique identification controls—significantly improving workflow efficiency for college administrators.

Skills

Languages: Python, SQL, C++, JavaScript

Databases: PostgreSQL, MySQL

Analytics & Visualization: Power BI, Excel, Pandas, NumPy, Matplotlib, Seaborn

Tools: Jupyter Notebook, VS Code, Git, GitHub, DataBricks

Education

Vellore Institute Of Technology

B.Tech, Computer Science & Engineering

CGPA: 8.45

Madhya Pradesh, India

May 2022 – July 2026

Lancers Army School, Surat

Class XII – CBSE

Percentage: 84.4%

Gujarat, India

Apr 2020 – Apr 2021

Lancers Army School, Surat

Class X – CBSE

Percentage: 94.4%

Gujarat, India

Apr 2018 – Apr 2019